

```
In [1]: !pip install py pandoc_binary
```

```
In [2]: import pandas as pd
```

```
In [3]: import numpy as np
```

Load Dataset

```
In [4]: df=pd.read_csv("C:/Users/sheheryarsh/Desktop/JUPYTER-LAB/datasets/employe
```

```
In [5]: print(df)
```

	Employee_ID	Name	Department	Hire_Date	Salary	\
0	E101	John Smith	Engineering	15/03/2021	85000	
1	E102	Jane Doe	Marketing	10/22/2022	65000	
2	E103	Mark Taylor	Sales	5/11/2023	72000	
3	E104	Sarah Connor	HR	12/1/2024	55000	
4	E105	David Lee	Engineering	1/15/2025	90000	
..	...	...	...	...	...	
95	E196	Tristan Knight	Engineering	14/09/2024	112000	
96	E197	Serenity Willis	Sales	18/02/2025	83000	
97	E198	Ryder Weaver	Marketing	9/24/2021	74000	
98	E199	Adeline Johnston	Legal	12/10/2022	96000	
99	E200	Sawyer Osborne	IT	21/11/2023	89000	

	Email
0	john.smith@company.com
1	jane.doe@company.com
2	mark.taylor@company.com
3	sarah.connor@company.com
4	david.lee@company.com
..	...
95	tristan.knight@company.com
96	serenity.willis@company.com
97	ryder.weaver@company.com
98	adeline.johnston@company.com
99	sawyer.osborne@company.com

[100 rows x 6 columns]

```
In [6]: df.head(6)
```

Out[6]:	Employee_ID	Name	Department	Hire_Date	Salary	Email
0	E101	John Smith	Engineering	15/03/2021	85000	john.smith@company.co
1	E102	Jane Doe	Marketing	10/22/2022	65000	jane.doe@company.co
2	E103	Mark Taylor	Sales	5/11/2023	72000	mark.taylor@company.co
3	E104	Sarah Connor	HR	12/1/2024	55000	sarah.connor@company.co
4	E105	David Lee	Engineering	1/15/2025	90000	david.lee@company.co
5	E101	John Smith	Engineering	15/03/2021	85000	john.smith@company.co

In [7]: `df.tail(7)`

Out[7]:	Employee_ID	Name	Department	Hire_Date	Salary	Email
93	E194	Ian Simpson	Operations	23/06/2022	54000	ian.simpson@comp
94	E188	Brody Medina	Engineering	11/12/2021	110000	brody.medina@comp
95	E196	Tristan Knight	Engineering	14/09/2024	112000	tristan.knight@comp
96	E197	Serenity Willis	Sales	18/02/2025	83000	serenity.willis@comp
97	E198	Ryder Weaver	Marketing	9/24/2021	74000	ryder.weaver@comp
98	E199	Adeline Johnston	Legal	12/10/2022	96000	adeline.johnston@comp
99	E200	Sawyer Osborne	IT	21/11/2023	89000	sawyer.osborne@comp

In [8]: `print(df.to_string())`

	Employee_ID	Name	Department	Hire_Date	Salary	
Email						
0	E101	John Smith	Engineering	15/03/2021	85000	
		john.smith@company.com				
1	E102	Jane Doe	Marketing	10/22/2022	65000	
		jane.doe@company.com				
2	E103	Mark Taylor	Sales	5/11/2023	72000	
		mark.taylor@company.com				
3	E104	Sarah Connor	HR	12/1/2024	55000	
		sarah.connor@company.com				
4	E105	David Lee	Engineering	1/15/2025	90000	
		david.lee@company.com				
5	E101	John Smith	Engineering	15/03/2021	85000	
		john.smith@company.com				
6	E107	Alice Brown	Finance	14/04/2022	95000	
7	E108	Robert Green	Operations	NaN	52000	
		robert.green@company.com				
8	E109	Emily White	IT	9/15/2023	seventy_k	
		emily.white@company.com				
9	E110	Michael Clark	Engineering	31/12/2021	110000	
		m.clark@company.com				
10	E111	Laura Hall	Marketing	2/8/2024	62000	
		laura.hall@company.com				
11	E112	James King	NaN	11/9/2025	81000	
		james.king@company.com				
12	E102	Jane Doe	Marketing	10/22/2022	65000	
		jane.doe@company.com				
13	E114	Emma Watson	Legal	15-05-2023	98000	
		emma.watson@company.com				
14	E115	Daniel Scott	HR	21/06/2022	58000	
		daniel.scott@company.com				
15	E116	Olivia Harris	Sales	10/1/2115	74000	
		olivia.harris@company.com				
16	E117	William Martin	IT	11/12/2023	88000	
		william.martin@company.com				
17	E118	Sophia Clark	Finance	17/02/2021	79000	
18	E119	Benjamin Lewis	Operations	19/04/2022	fifty_k	
		ben.lewis@company.com				
19	E120	Charlotte Walker	NaN	24/09/2023	82000	ch
		arlotte.walker@company.com				
20	E115	Daniel Scott	HR	21/06/2022	58000	
		daniel.scott@company.com				
21	E122	Alexander Young	Engineering	12/10/2024	94000	
		alex.young@company.com				
22	E123	Isabella King	Sales	5/3/2025	71000	
		isabella.king@company.com				
23	E124	Lucas Wright	Marketing	1/14/2022	63000	
		lucas.wright@company.com				
24	E125	Amelia Hill	Legal	16/08/2021	105000	
		amelia.hill@company.com				
25	E126	Mason Allen	IT	19-11-2023	89000	
		mason.allen@company.com				
26	E127	Evelyn Baker	Finance	2/12/2024	93000	
		evelyn.baker@company.com				
27	E128	Logan Carter	Operations	27/05/2022	48000	
28	E129	Harper Mitchell	HR	8/10/2025	59000	h
		arper.mitchell@company.com				

29	E130	Jackson Perez	Engineering	12/4/2100	102000	
		jackson.perez@company.com				
30	E131	Abigail Roberts	Sales	15/09/2023	73000	a
		bigail.roberts@company.com				
31	E132	Sebastian Turner	Marketing	4/11/2021	68000	
		seb.turner@company.com				
32	E133	Avery Phillips	Legal	18/01/2022	91000	
		avery.phillips@company.com				
33	E134	Mia Campbell	Engineering	10/25/2024	99000	
		mia.campbell@company.com				
34	E135	Ethan Parker	Finance	14/03/2025	97000	
		ethan.parker@company.com				
35	E136	Ella Evans	IT	21/07/2022	86000	
		ella.evans@company.com				
36	E137	Jacob Edwards	Operations	9/12/2023	54000	
		jacob.edwards@company.com				
37	E138	Sofia Collins	HR	17/04/2021	60000	
		sofia.collins@company.com				
38	E139	Jackson Stewart	Engineering	13/11/2022	108000	j
		ackson.stewart@company.com				
39	E140	Aria Morris	Sales	12/5/2024	76000	
		aria.morris@company.com				
40	E137	Jacob Edwards	Operations	9/12/2023	54000	
		jacob.edwards@company.com				
41	E142	Logan Rogers	Marketing	14/02/2021	66000	
		logan.rogers@company.com				
42	E143	Camila Reed	Legal	8/16/2022	92000	
		camila.reed@company.com				
43	E144	Benjamin Cook	IT	11/9/2023	87000	
		ben.cook@company.com				
44	E145	Chloe Morgan	Finance	25/04/2024	96000	
		chloe.morgan@company.com				
45	E146	Oliver Bell	Operations	16-10-2025	49000	
		NaN				
46	E147	Victoria Murphy	HR	19/01/2022	61000	v
		ictoria.murphy@company.com				
47	E148	Samuel Bailey	Engineering	2/28/2023	101000	
		samuel.bailey@company.com				
48	E149	Zoey Rivera	Sales	13/06/2024	77000	
		zoey.rivera@company.com				
49	E150	Gabriel Cooper	Marketing	17/07/2025	69000	
		gabriel.cooper@company.com				
50	E151	Lily Richardson	Legal	10/15/2021	88000	l
		ily.richardson@company.com				
51	E152	Carter Cox	IT	22/11/2022	90000	
		carter.cox@company.com				
52	E153	Hannah Howard	Finance	7/9/2023	94000	
		hannah.howard@company.com				
53	E154	Luke Ward	Operations	15/10/2024	51000	
		luke.ward@company.com				
54	E155	Addison Torres	HR	18/02/2021	62000	
		addison.torres@company.com				
55	E156	Isaac Peterson	Engineering	11/12/2022	104000	
		isaac.peterson@company.com				
56	E157	Nora Gray	Sales	23/05/2023	78000	
		nora.gray@company.com				
57	E158	Julian Ramirez	Marketing	12/4/2024	67000	
		julian.ramirez@company.com				
58	E159	Grace James	Legal	14/06/2025	93000	
		grace.james@company.com				

59	E160	Levi Jenkins	IT	8/27/2021	84000	
levi.jenkins@company.com						
60	E161	Hazel Diaz	Finance	19/09/2022	95000	
hazel.diaz@company.com						
61	E162	Ryan Hayes	Operations	10/12/2023	47000	
ryan.hayes@company.com						
62	E163	Eleanor Myers	HR	14-02-2024	63000	
eleanor.myers@company.com						
63	E164	Nathan Washington	Engineering	25/04/2025	106000	nat
han.washington@company.com						
64	E165	Penelope Foster	Sales	3/11/2021	79000	p
enelope.foster@company.com						
65	E166	Aaron Butler	Marketing	12/7/2022	70000	
aaron.butler@company.com						
66	E155	Addison Torres	HR	18/02/2021	62000	
addison.torres@company.com						
67	E168	Connor Simmons	IT	10/18/2024	82000	
connor.simmons@company.com						
68	E169	Stella Hughes	Finance	21/09/2025	98000	
stella.hughes@company.com						
69	E170	Adrian Henderson	Operations	5/1/2022	53000	
NaN						
70	E171	Paisley Alvarez	HR	13/02/2023	64000	p
aisley.alvarez@company.com						
71	E172	Hudson Powell	Engineering	10/3/2024	107000	
hudson.powell@company.com						
72	E173	Elena Fernandez	Sales	14-04-2025	80000	e
lena.fernandez@company.com						
73	E174	Liam Pierce	Marketing	22/05/2021	71000	
liam.pierce@company.com						
74	E175	Maya Reynolds	Legal	9/28/2022	91000	
maya.reynolds@company.com						
75	E176	Leo Griffin	IT	16/10/2023	85000	
leo.griffin@company.com						
76	E177	Mackenzie Hunt	Finance	12/11/2024	99000	
mackenzie.hunt@company.com						
77	E178	Eli Hunter	Operations	10/9/2021	50000	
eli.hunter@company.com						
78	E179	Savannah Tucker	HR	14/01/2022	65000	s
avannah.tucker@company.com						
79	E180	Asher Hicks	Engineering	25/02/2023	109000	
asher.hicks@company.com						
80	E181	Audrey Crawford	Sales	9/12/2024	81000	a
udrey.crawford@company.com						
81	E182	Miles Ortiz	Marketing	13/03/2025	72000	
miles.ortiz@company.com						
82	E183	Claire Chapman	Legal	17/04/2021	94000	
claire.chapman@company.com						
83	E184	Nolan Romero	IT	10/21/2022	86000	
nolan.romero@company.com						
84	E185	Peyton Hawkins	Finance	14/05/2023	97000	
peyton.hawkins@company.com						
85	E186	Camden Moreno	Operations	22/06/2024	52000	
camden.moreno@company.com						
86	E187	Kaylee Fox	HR	9/7/2025	66000	
kaylee.fox@company.com						
87	E188	Brody Medina	Engineering	11/12/2021	110000	
brody.medina@company.com						
88	E189	Allison Alvarez	Sales	15/01/2022	82000	a
llison.alvarez@company.com						

89	E190	Julian Caldwell	Marketing	19-02-2023	73000	
NaN						
90	E191	Josephine Hansen	Legal	26/03/2024	95000	jo
sephine.hansen@company.com						
91	E192	Lincoln Matthews	IT	10/25/2025	87000	li
ncoln.matthews@company.com						
92	E193	Eva Gutierrez	Finance	14/05/2021	98000	
eva.gutierrez@company.com						
93	E194	Ian Simpson	Operations	23/06/2022	54000	
ian.simpson@company.com						
94	E188	Brody Medina	Engineering	11/12/2021	110000	
brody.medina@company.com						
95	E196	Tristan Knight	Engineering	14/09/2024	112000	
tristan.knight@company.com						
96	E197	Serenity Willis	Sales	18/02/2025	83000	s
erenity.willis@company.com						
97	E198	Ryder Weaver	Marketing	9/24/2021	74000	
ryder.weaver@company.com						
98	E199	Adeline Johnston	Legal	12/10/2022	96000	ad
eline.johnston@company.com						
99	E200	Sawyer Osborne	IT	21/11/2023	89000	
sawyer.osborne@company.com						

In [9]: `df.loc[0]`

Out[9]:

Employee_ID	E101
Name	John Smith
Department	Engineering
Hire_Date	15/03/2021
Salary	85000
Email	john.smith@company.com

Name: 0, dtype: str

In [10]: `df.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Employee_ID     100 non-null   str
1   Name            100 non-null   str
2   Department       98 non-null    str
3   Hire_Date       99 non-null    str
4   Salary          100 non-null   str
5   Email           94 non-null    str
dtypes: str(6)
memory usage: 4.8 KB
```

Total number of missing values per Column

In [11]: `df.isna().sum()`

Out[11]:

Employee_ID	0
Name	0
Department	2
Hire_Date	1
Salary	0
Email	6

dtype: int64

```
In [12]: df[df['Hire_Date'].isna()]
```

```
Out[12]:
```

	Employee_ID	Name	Department	Hire_Date	Salary	Email
7	E108	Robert Green	Operations	NaN	52000	robert.green@company.com

```
In [13]: df[df['Department'].isna()]
```

```
Out[13]:
```

	Employee_ID	Name	Department	Hire_Date	Salary	
11	E112	James King	NaN	11/9/2025	81000	james.king@compa
19	E120	Charlotte Walker	NaN	24/09/2023	82000	charlotte.walker@compa

Removing Duplicates

```
In [14]: df.drop_duplicates(keep='first',inplace=True)
```

```
In [15]: print(df.to_string())
```

	Employee_ID	Name	Department	Hire_Date	Salary	
Email						
0	E101	John Smith	Engineering	15/03/2021	85000	
		john.smith@company.com				
1	E102	Jane Doe	Marketing	10/22/2022	65000	
		jane.doe@company.com				
2	E103	Mark Taylor	Sales	5/11/2023	72000	
		mark.taylor@company.com				
3	E104	Sarah Connor	HR	12/1/2024	55000	
		sarah.connor@company.com				
4	E105	David Lee	Engineering	1/15/2025	90000	
		david.lee@company.com				
6	E107	Alice Brown	Finance	14/04/2022	95000	
		NaN				
7	E108	Robert Green	Operations	NaN	52000	
		robert.green@company.com				
8	E109	Emily White	IT	9/15/2023	seventy_k	
		emily.white@company.com				
9	E110	Michael Clark	Engineering	31/12/2021	110000	
		m.clark@company.com				
10	E111	Laura Hall	Marketing	2/8/2024	62000	
		laura.hall@company.com				
11	E112	James King	NaN	11/9/2025	81000	
		james.king@company.com				
13	E114	Emma Watson	Legal	15-05-2023	98000	
		emma.watson@company.com				
14	E115	Daniel Scott	HR	21/06/2022	58000	
		daniel.scott@company.com				
15	E116	Olivia Harris	Sales	10/1/2115	74000	
		olivia.harris@company.com				
16	E117	William Martin	IT	11/12/2023	88000	
		william.martin@company.com				
17	E118	Sophia Clark	Finance	17/02/2021	79000	
		NaN				
18	E119	Benjamin Lewis	Operations	19/04/2022	fifty_k	
		ben.lewis@company.com				
19	E120	Charlotte Walker	NaN	24/09/2023	82000	ch
		arlotte.walker@company.com				
21	E122	Alexander Young	Engineering	12/10/2024	94000	
		alex.young@company.com				
22	E123	Isabella King	Sales	5/3/2025	71000	
		isabella.king@company.com				
23	E124	Lucas Wright	Marketing	1/14/2022	63000	
		lucas.wright@company.com				
24	E125	Amelia Hill	Legal	16/08/2021	105000	
		amelia.hill@company.com				
25	E126	Mason Allen	IT	19-11-2023	89000	
		mason.allen@company.com				
26	E127	Evelyn Baker	Finance	2/12/2024	93000	
		evelyn.baker@company.com				
27	E128	Logan Carter	Operations	27/05/2022	48000	
		NaN				
28	E129	Harper Mitchell	HR	8/10/2025	59000	h
		arper.mitchell@company.com				
29	E130	Jackson Perez	Engineering	12/4/2100	102000	
		jackson.perez@company.com				
30	E131	Abigail Roberts	Sales	15/09/2023	73000	a
		bigail.roberts@company.com				
31	E132	Sebastian Turner	Marketing	4/11/2021	68000	
		seb.turner@company.com				

32	E133	Avery Phillips	Legal	18/01/2022	91000	
		avery.phillips@company.com				
33	E134	Mia Campbell	Engineering	10/25/2024	99000	
		mia.campbell@company.com				
34	E135	Ethan Parker	Finance	14/03/2025	97000	
		ethan.parker@company.com				
35	E136	Ella Evans	IT	21/07/2022	86000	
		ella.evans@company.com				
36	E137	Jacob Edwards	Operations	9/12/2023	54000	
		jacob.edwards@company.com				
37	E138	Sofia Collins	HR	17/04/2021	60000	
		sofia.collins@company.com				
38	E139	Jackson Stewart	Engineering	13/11/2022	108000	j
		ackson.stewart@company.com				
39	E140	Aria Morris	Sales	12/5/2024	76000	
		aria.morris@company.com				
41	E142	Logan Rogers	Marketing	14/02/2021	66000	
		logan.rogers@company.com				
42	E143	Camila Reed	Legal	8/16/2022	92000	
		camila.reed@company.com				
43	E144	Benjamin Cook	IT	11/9/2023	87000	
		ben.cook@company.com				
44	E145	Chloe Morgan	Finance	25/04/2024	96000	
		chloe.morgan@company.com				
45	E146	Oliver Bell	Operations	16-10-2025	49000	
		NaN				
46	E147	Victoria Murphy	HR	19/01/2022	61000	v
		ictoria.murphy@company.com				
47	E148	Samuel Bailey	Engineering	2/28/2023	101000	
		samuel.bailey@company.com				
48	E149	Zoey Rivera	Sales	13/06/2024	77000	
		zoey.rivera@company.com				
49	E150	Gabriel Cooper	Marketing	17/07/2025	69000	
		gabriel.cooper@company.com				
50	E151	Lily Richardson	Legal	10/15/2021	88000	l
		ily.richardson@company.com				
51	E152	Carter Cox	IT	22/11/2022	90000	
		carter.cox@company.com				
52	E153	Hannah Howard	Finance	7/9/2023	94000	
		hannah.howard@company.com				
53	E154	Luke Ward	Operations	15/10/2024	51000	
		luke.ward@company.com				
54	E155	Addison Torres	HR	18/02/2021	62000	
		addison.torres@company.com				
55	E156	Isaac Peterson	Engineering	11/12/2022	104000	
		isaac.peterson@company.com				
56	E157	Nora Gray	Sales	23/05/2023	78000	
		nora.gray@company.com				
57	E158	Julian Ramirez	Marketing	12/4/2024	67000	
		julian.ramirez@company.com				
58	E159	Grace James	Legal	14/06/2025	93000	
		grace.james@company.com				
59	E160	Levi Jenkins	IT	8/27/2021	84000	
		levi.jenkins@company.com				
60	E161	Hazel Diaz	Finance	19/09/2022	95000	
		hazel.diaz@company.com				
61	E162	Ryan Hayes	Operations	10/12/2023	47000	
		ryan.hayes@company.com				
62	E163	Eleanor Myers	HR	14-02-2024	63000	
		eleanor.myers@company.com				

63	E164	Nathan Washington	Engineering	25/04/2025	106000	nat
han.washington@company.com						
64	E165	Penelope Foster	Sales	3/11/2021	79000	p
enelope.foster@company.com						
65	E166	Aaron Butler	Marketing	12/7/2022	70000	
aaron.butler@company.com						
67	E168	Connor Simmons	IT	10/18/2024	82000	
connor.simmons@company.com						
68	E169	Stella Hughes	Finance	21/09/2025	98000	
stella.hughes@company.com						
69	E170	Adrian Henderson	Operations	5/1/2022	53000	
NaN						
70	E171	Paisley Alvarez	HR	13/02/2023	64000	p
aisley.alvarez@company.com						
71	E172	Hudson Powell	Engineering	10/3/2024	107000	
hudson.powell@company.com						
72	E173	Elena Fernandez	Sales	14-04-2025	80000	e
lena.fernandez@company.com						
73	E174	Liam Pierce	Marketing	22/05/2021	71000	
liam.pierce@company.com						
74	E175	Maya Reynolds	Legal	9/28/2022	91000	
maya.reynolds@company.com						
75	E176	Leo Griffin	IT	16/10/2023	85000	
leo.griffin@company.com						
76	E177	Mackenzie Hunt	Finance	12/11/2024	99000	
mackenzie.hunt@company.com						
77	E178	Eli Hunter	Operations	10/9/2021	50000	
eli.hunter@company.com						
78	E179	Savannah Tucker	HR	14/01/2022	65000	s
avannah.tucker@company.com						
79	E180	Asher Hicks	Engineering	25/02/2023	109000	
asher.hicks@company.com						
80	E181	Audrey Crawford	Sales	9/12/2024	81000	a
udrey.crawford@company.com						
81	E182	Miles Ortiz	Marketing	13/03/2025	72000	
miles.ortiz@company.com						
82	E183	Claire Chapman	Legal	17/04/2021	94000	
claire.chapman@company.com						
83	E184	Nolan Romero	IT	10/21/2022	86000	
nolan.romero@company.com						
84	E185	Peyton Hawkins	Finance	14/05/2023	97000	
peyton.hawkins@company.com						
85	E186	Camden Moreno	Operations	22/06/2024	52000	
camden.moreno@company.com						
86	E187	Kaylee Fox	HR	9/7/2025	66000	
kaylee.fox@company.com						
87	E188	Brody Medina	Engineering	11/12/2021	110000	
brody.medina@company.com						
88	E189	Allison Alvarez	Sales	15/01/2022	82000	a
llison.alvarez@company.com						
89	E190	Julian Caldwell	Marketing	19-02-2023	73000	
NaN						
90	E191	Josephine Hansen	Legal	26/03/2024	95000	jo
sephine.hansen@company.com						
91	E192	Lincoln Matthews	IT	10/25/2025	87000	li
ncoln.matthews@company.com						
92	E193	Eva Gutierrez	Finance	14/05/2021	98000	
eva.gutierrez@company.com						
93	E194	Ian Simpson	Operations	23/06/2022	54000	
ian.simpson@company.com						

95	E196	Tristan Knight	Engineering	14/09/2024	112000	
tristan.knight@company.com						
96	E197	Serenity Willis	Sales	18/02/2025	83000	s
erenity.willis@company.com						
97	E198	Ryder Weaver	Marketing	9/24/2021	74000	
ryder.weaver@company.com						
98	E199	Adeline Johnston	Legal	12/10/2022	96000	ad
eline.johnston@company.com						
99	E200	Sawyer Osborne	IT	21/11/2023	89000	
sawyer.osborne@company.com						

In [16]: `df.duplicated()`

Out[16]:

0	False
1	False
2	False
3	False
4	False
...	
95	False
96	False
97	False
98	False
99	False

Length: 94, dtype: bool

Correct wrong Salary data & Convert into Numeric

In [17]: `print(df['Salary'].to_string())`

0	85000
1	65000
2	72000
3	55000
4	90000
6	95000
7	52000
8	seventy_k
9	110000
10	62000
11	81000
13	98000
14	58000
15	74000
16	88000
17	79000
18	fifty_k
19	82000
21	94000
22	71000
23	63000
24	105000
25	89000
26	93000
27	48000
28	59000
29	102000
30	73000
31	68000
32	91000
33	99000
34	97000
35	86000
36	54000
37	60000
38	108000
39	76000
41	66000
42	92000
43	87000
44	96000
45	49000
46	61000
47	101000
48	77000
49	69000
50	88000
51	90000
52	94000
53	51000
54	62000
55	104000
56	78000
57	67000
58	93000
59	84000
60	95000
61	47000
62	63000
63	106000

64	79000
65	70000
67	82000
68	98000
69	53000
70	64000
71	107000
72	80000
73	71000
74	91000
75	85000
76	99000
77	50000
78	65000
79	109000
80	81000
81	72000
82	94000
83	86000
84	97000
85	52000
86	66000
87	110000
88	82000
89	73000
90	95000
91	87000
92	98000
93	54000
95	112000
96	83000
97	74000
98	96000
99	89000

```
In [18]: sal_mapping={'seventy_k':70000, 'fifty_k':50000}  
df['Salary']=df['Salary'].replace(sal_mapping)
```

```
In [19]: print(df['Salary'].to_string())
```

0	85000
1	65000
2	72000
3	55000
4	90000
6	95000
7	52000
8	70000
9	110000
10	62000
11	81000
13	98000
14	58000
15	74000
16	88000
17	79000
18	50000
19	82000
21	94000
22	71000
23	63000
24	105000
25	89000
26	93000
27	48000
28	59000
29	102000
30	73000
31	68000
32	91000
33	99000
34	97000
35	86000
36	54000
37	60000
38	108000
39	76000
41	66000
42	92000
43	87000
44	96000
45	49000
46	61000
47	101000
48	77000
49	69000
50	88000
51	90000
52	94000
53	51000
54	62000
55	104000
56	78000
57	67000
58	93000
59	84000
60	95000
61	47000
62	63000
63	106000

64	79000
65	70000
67	82000
68	98000
69	53000
70	64000
71	107000
72	80000
73	71000
74	91000
75	85000
76	99000
77	50000
78	65000
79	109000
80	81000
81	72000
82	94000
83	86000
84	97000
85	52000
86	66000
87	110000
88	82000
89	73000
90	95000
91	87000
92	98000
93	54000
95	112000
96	83000
97	74000
98	96000
99	89000

```
In [20]: df['Salary']=pd.to_numeric(df['Salary'],errors='coerce')
```

```
In [21]: print(df['Hire_Date'].to_string())
```

0	15/03/2021
1	10/22/2022
2	5/11/2023
3	12/1/2024
4	1/15/2025
6	14/04/2022
7	NaN
8	9/15/2023
9	31/12/2021
10	2/8/2024
11	11/9/2025
13	15-05-2023
14	21/06/2022
15	10/1/2115
16	11/12/2023
17	17/02/2021
18	19/04/2022
19	24/09/2023
21	12/10/2024
22	5/3/2025
23	1/14/2022
24	16/08/2021
25	19-11-2023
26	2/12/2024
27	27/05/2022
28	8/10/2025
29	12/4/2100
30	15/09/2023
31	4/11/2021
32	18/01/2022
33	10/25/2024
34	14/03/2025
35	21/07/2022
36	9/12/2023
37	17/04/2021
38	13/11/2022
39	12/5/2024
41	14/02/2021
42	8/16/2022
43	11/9/2023
44	25/04/2024
45	16-10-2025
46	19/01/2022
47	2/28/2023
48	13/06/2024
49	17/07/2025
50	10/15/2021
51	22/11/2022
52	7/9/2023
53	15/10/2024
54	18/02/2021
55	11/12/2022
56	23/05/2023
57	12/4/2024
58	14/06/2025
59	8/27/2021
60	19/09/2022
61	10/12/2023
62	14-02-2024
63	25/04/2025

64	3/11/2021
65	12/7/2022
67	10/18/2024
68	21/09/2025
69	5/1/2022
70	13/02/2023
71	10/3/2024
72	14-04-2025
73	22/05/2021
74	9/28/2022
75	16/10/2023
76	12/11/2024
77	10/9/2021
78	14/01/2022
79	25/02/2023
80	9/12/2024
81	13/03/2025
82	17/04/2021
83	10/21/2022
84	14/05/2023
85	22/06/2024
86	9/7/2025
87	11/12/2021
88	15/01/2022
89	19-02-2023
90	26/03/2024
91	10/25/2025
92	14/05/2021
93	23/06/2022
95	14/09/2024
96	18/02/2025
97	9/24/2021
98	12/10/2022
99	21/11/2023

```
In [22]: df['Hire_Date'] = pd.to_datetime(df['Hire_Date'], format='mixed', errors=''
```

```
In [23]: print(df['Hire_Date'].to_string())
```

0	2021-03-15
1	2022-10-22
2	2023-05-11
3	2024-12-01
4	2025-01-15
6	2022-04-14
7	NaT
8	2023-09-15
9	2021-12-31
10	2024-02-08
11	2025-11-09
13	2023-05-15
14	2022-06-21
15	2115-10-01
16	2023-11-12
17	2021-02-17
18	2022-04-19
19	2023-09-24
21	2024-12-10
22	2025-05-03
23	2022-01-14
24	2021-08-16
25	2023-11-19
26	2024-02-12
27	2022-05-27
28	2025-08-10
29	2100-12-04
30	2023-09-15
31	2021-04-11
32	2022-01-18
33	2024-10-25
34	2025-03-14
35	2022-07-21
36	2023-09-12
37	2021-04-17
38	2022-11-13
39	2024-12-05
41	2021-02-14
42	2022-08-16
43	2023-11-09
44	2024-04-25
45	2025-10-16
46	2022-01-19
47	2023-02-28
48	2024-06-13
49	2025-07-17
50	2021-10-15
51	2022-11-22
52	2023-07-09
53	2024-10-15
54	2021-02-18
55	2022-11-12
56	2023-05-23
57	2024-12-04
58	2025-06-14
59	2021-08-27
60	2022-09-19
61	2023-10-12
62	2024-02-14
63	2025-04-25

```
64 2021-03-11
65 2022-12-07
67 2024-10-18
68 2025-09-21
69 2022-05-01
70 2023-02-13
71 2024-10-03
72 2025-04-14
73 2021-05-22
74 2022-09-28
75 2023-10-16
76 2024-12-11
77 2021-10-09
78 2022-01-14
79 2023-02-25
80 2024-09-12
81 2025-03-13
82 2021-04-17
83 2022-10-21
84 2023-05-14
85 2024-06-22
86 2025-09-07
87 2021-11-12
88 2022-01-15
89 2023-02-19
90 2024-03-26
91 2025-10-25
92 2021-05-14
93 2022-06-23
95 2024-09-14
96 2025-02-18
97 2021-09-24
98 2022-12-10
99 2023-11-21
```

```
In [24]: df['Hire_Date'] = df['Hire_Date'].dt.strftime('%m-%d-%Y')
```

```
In [25]: print(df['Hire_Date'].to_string())
```

0	03-15-2021
1	10-22-2022
2	05-11-2023
3	12-01-2024
4	01-15-2025
6	04-14-2022
7	NaN
8	09-15-2023
9	12-31-2021
10	02-08-2024
11	11-09-2025
13	05-15-2023
14	06-21-2022
15	10-01-2115
16	11-12-2023
17	02-17-2021
18	04-19-2022
19	09-24-2023
21	12-10-2024
22	05-03-2025
23	01-14-2022
24	08-16-2021
25	11-19-2023
26	02-12-2024
27	05-27-2022
28	08-10-2025
29	12-04-2100
30	09-15-2023
31	04-11-2021
32	01-18-2022
33	10-25-2024
34	03-14-2025
35	07-21-2022
36	09-12-2023
37	04-17-2021
38	11-13-2022
39	12-05-2024
41	02-14-2021
42	08-16-2022
43	11-09-2023
44	04-25-2024
45	10-16-2025
46	01-19-2022
47	02-28-2023
48	06-13-2024
49	07-17-2025
50	10-15-2021
51	11-22-2022
52	07-09-2023
53	10-15-2024
54	02-18-2021
55	11-12-2022
56	05-23-2023
57	12-04-2024
58	06-14-2025
59	08-27-2021
60	09-19-2022
61	10-12-2023
62	02-14-2024
63	04-25-2025

64	03-11-2021
65	12-07-2022
67	10-18-2024
68	09-21-2025
69	05-01-2022
70	02-13-2023
71	10-03-2024
72	04-14-2025
73	05-22-2021
74	09-28-2022
75	10-16-2023
76	12-11-2024
77	10-09-2021
78	01-14-2022
79	02-25-2023
80	09-12-2024
81	03-13-2025
82	04-17-2021
83	10-21-2022
84	05-14-2023
85	06-22-2024
86	09-07-2025
87	11-12-2021
88	01-15-2022
89	02-19-2023
90	03-26-2024
91	10-25-2025
92	05-14-2021
93	06-23-2022
95	09-14-2024
96	02-18-2025
97	09-24-2021
98	12-10-2022
99	11-21-2023

```
In [26]: print(df.shape)
```

```
(94, 6)
```

Fill all NaN values across all columns with 'Unknown'

```
In [27]: df = df.fillna('Unknown')
```

```
In [28]: print(df['Department'].to_string())
```

0	Engineering
1	Marketing
2	Sales
3	HR
4	Engineering
6	Finance
7	Operations
8	IT
9	Engineering
10	Marketing
11	Unknown
13	Legal
14	HR
15	Sales
16	IT
17	Finance
18	Operations
19	Unknown
21	Engineering
22	Sales
23	Marketing
24	Legal
25	IT
26	Finance
27	Operations
28	HR
29	Engineering
30	Sales
31	Marketing
32	Legal
33	Engineering
34	Finance
35	IT
36	Operations
37	HR
38	Engineering
39	Sales
41	Marketing
42	Legal
43	IT
44	Finance
45	Operations
46	HR
47	Engineering
48	Sales
49	Marketing
50	Legal
51	IT
52	Finance
53	Operations
54	HR
55	Engineering
56	Sales
57	Marketing
58	Legal
59	IT
60	Finance
61	Operations
62	HR
63	Engineering

```
64         Sales
65     Marketing
66         IT
67         Finance
68     Operations
69         HR
70     Engineering
71         Sales
72     Marketing
73         Legal
74         IT
75         Finance
76     Operations
77         HR
78     Engineering
79         Sales
80     Marketing
81         Legal
82         IT
83         Finance
84     Operations
85         HR
86     Engineering
87         Sales
88     Marketing
89         Legal
90         IT
91         Finance
92     Operations
93     Engineering
94         Sales
95     Marketing
96         Legal
97         IT
```

```
In [29]: df.isna().sum()
```

```
Out[29]: Employee_ID    0
         Name           0
         Department     0
         Hire_Date      0
         Salary         0
         Email          0
         dtype: int64
```